

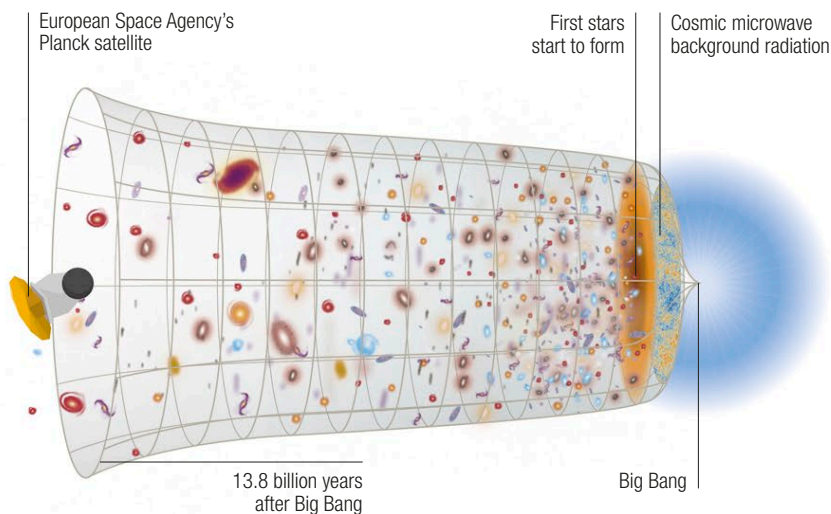
SIZE AND STRUCTURE OF THE UNIVERSE

THE EXTENT OF THE UNIVERSE WE CAN SEE AROUND US IS LIMITED BY THE SPEED OF LIGHT AND THE RATE OF COSMIC EXPANSION, BUT THE UNIVERSE AS A WHOLE GOES FAR BEYOND THESE LIMITS.

The Big Bang explosion that created the Universe some 13.8 billion years ago produced not just matter, but also space and time. As a result, there is a limit to how far we can theoretically see across space because we can only see regions whose light has had time to reach us. This places us at the center of a spherical bubble called our observable Universe, but space itself extends far beyond this boundary. In fact, every location in the cosmos is at the center of an observable Universe of its own.

Cosmic microwave background radiation

Immediately after the Big Bang, the Universe was an opaque, expanding fireball from which no light could escape. As a result, the most distant region of the cosmos that we can actually see corresponds to the period about 380,000 years later, when the “fog” of the early Universe cleared. Light from the edge of the fireball can still be seen if we look far enough in any direction, but during its 13.8-billion-year journey to reach us it has become redshifted (see p.70) so that the fireball’s light now makes the whole sky glow at microwave wavelengths.

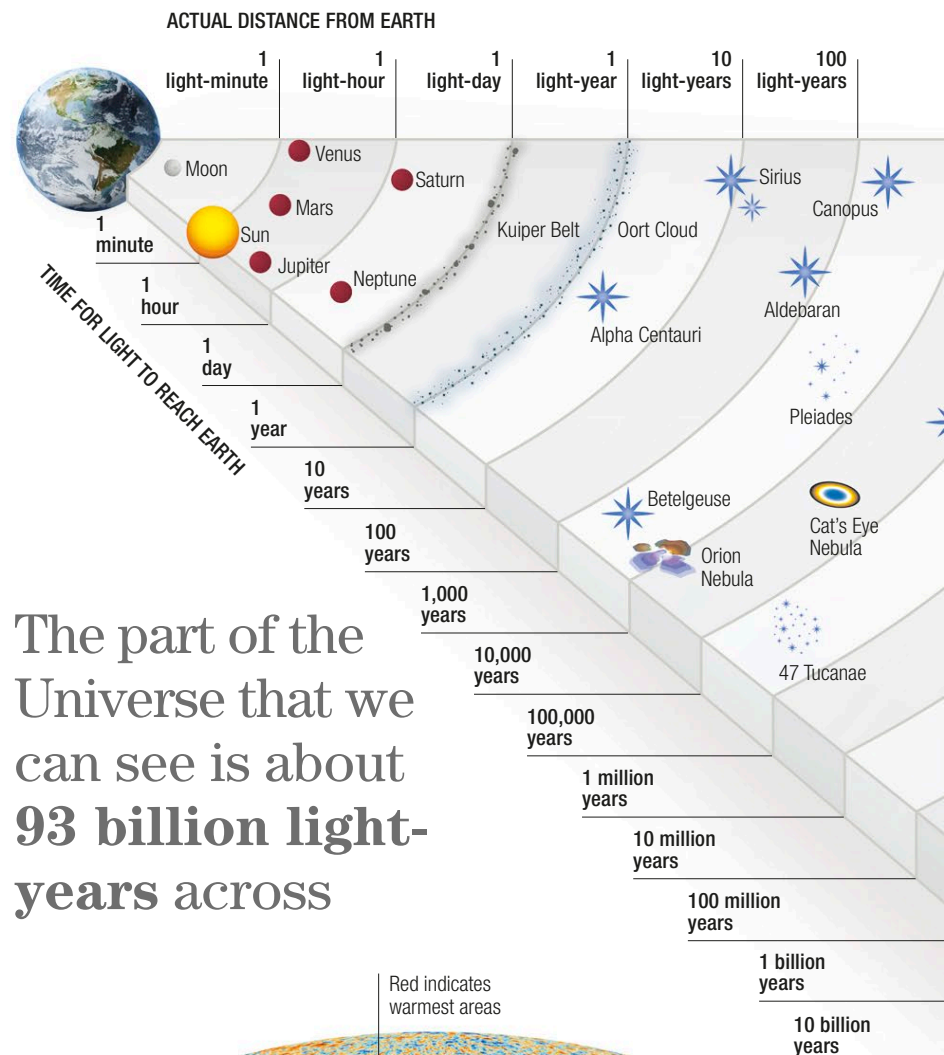


△ Looking back in time

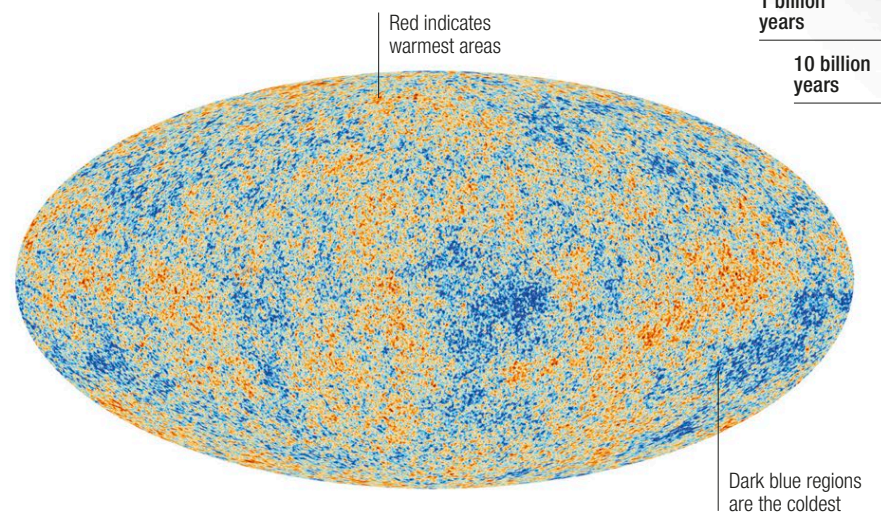
Looking farther away in space is one way of studying features of the young Universe that do not survive today. At great distances, we can see young galaxies busy forming, and soon we may even see radiation from the very first stars. The cosmic microwave background marks the earliest and most distant light radiation we can detect.

Observable Universe

Our observations of the Universe are limited to those objects whose light has had time to reach us over the past 13.8 billion years. However, thanks to cosmic expansion (see pp.70–71), the farthest regions of the Universe are moving away from us at the speed of light itself. As a result, light from regions beyond the observable Universe will never be able to reach Earth, no matter how much time passes.



The part of the Universe that we can see is about **93 billion light-years** across



△ Radiation map

The European Space Agency's Planck satellite mapped variations in the microwave background radiation corresponding to infinitesimal temperature differences. These reveal tiny variations in the temperature and density of the earlier Universe, showing where structures were starting to form even at this early time.